

LAZARD LCOE

LEVELIZED COST OF ENERGY+

June 2025

WITH SUPPORT FROM



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Executive Summary

Executive Summary—Selected Key Findings from Lazard’s 2025 LCOE+

Lazard’s 2025 LCOE+ Report is organized around three key areas: Energy Generation, Energy Storage and the Energy System

Energy Generation

Levelized Cost of Energy Version 18.0

- **Renewables Remain Competitive:** On an unsubsidized \$/MWh basis, renewable energy remains the most cost-competitive form of generation. As such, renewable energy will continue to play a key role in the buildout of new power generation in the U.S. This is particularly true in the current high power demand environment, where renewables stand out as both the lowest-cost and quickest-to-deploy generation resource
- **Increasing Competitiveness of Existing Gas Generation:** The gap between the LCOE of new wind and solar and the marginal cost of operating CCGTs has widened due to, among other things, persistent low gas prices, high energy demand and increasing renewable LCOEs
- **Significant Shifts Expected:** Unless otherwise indicated, Lazard’s LCOE is an LTM analysis focused on “today” and is not a forecasting tool. As such, the outcomes included herein are representative of current development and construction timelines, which vary by technology. For example, while this year’s analysis shows only a slight increase in the LCOE of CCGTs, turbine shortages, rising costs and long lead times are expected to drive steep LCOE increases for gas technologies in the near term, as illustrated herein. Additionally, cost declines across Vogtle units 3 and 4 indicate nuclear is poised to benefit from scale and development efficiencies

Energy Storage

Levelized Cost of Storage Version 10.0

- **Storage Cost Decline:** This year’s analysis shows notable declines in the LCOS of utility scale and C&I battery energy storage systems. Key drivers of such results include both market dynamics (e.g., lower-than-expected EV demand and the resulting oversupply of cells) and technological advancements (e.g., increased cell capacity and energy density)
- **Tariffs Increase Uncertainty:** While current pricing is further benefiting from aggressive competition, widening LCOS spreads indicate increased volatility as uncertainty related to the ultimate tariff regime is shaping market dynamics in real time. For example, supply chain relocation to Southeast Asia and India is well underway, and market participants are executing on forward procurement strategies to mitigate future pricing risk
- **Market Expansion Is Underway:** The LCOS value snapshots show increased returns reflecting the confluence of lower costs and higher prices in several regions. Energy storage adoption is expanding beyond ISO/RTO-driven wholesale markets and into states where municipal procurement and data center growth is prevalent (e.g., Arizona, Colorado, Florida). Lazard expects continued expansion as backup power and grid resilience become increasingly important in high-growth markets

Energy System

Cost of Firming Intermittency

- **Firming Value Rises as Renewable Penetration Increases:** The cost of firming helps grid operators evaluate resources based on a region’s existing generation mix and load characteristics, ensuring the right balance between reliability and affordability. The results of this year’s firming analysis show that as the penetration of low-cost intermittent generation increases, the value of firm capacity rises
- **ISO Approaches to System Analysis Are Evolving:** Several independent system operators are adjusting their capacity accreditation methodologies in ways that are generally increasing firming costs. Both CAISO and PJM have reduced capacity accreditation values for highly correlated resources (e.g., solar and shorter-duration storage). Continued development of more sophisticated capacity accreditation frameworks, such as incorporation of seasonal adjustments or diversity benefits, could have material impacts on future firming costs
- **Diverse Generation Sources and Innovation Are Needed:** The results of Lazard’s LCOE+ have consistently supported deploying a diverse mix of energy resources. Despite the sustained unsubsidized cost competitiveness of renewable energy, resource planning metrics indicate diverse generation fleets will be required over the long term to meet power needs, likely bolstered by now-emerging technologies such as long duration energy storage, geothermal, nuclear small modular reactors, pumped storage hydropower and carbon capture and storage, among others



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Energy Generation



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Lazard's Levelized Cost of Energy Analysis—Version 18.0

Introduction

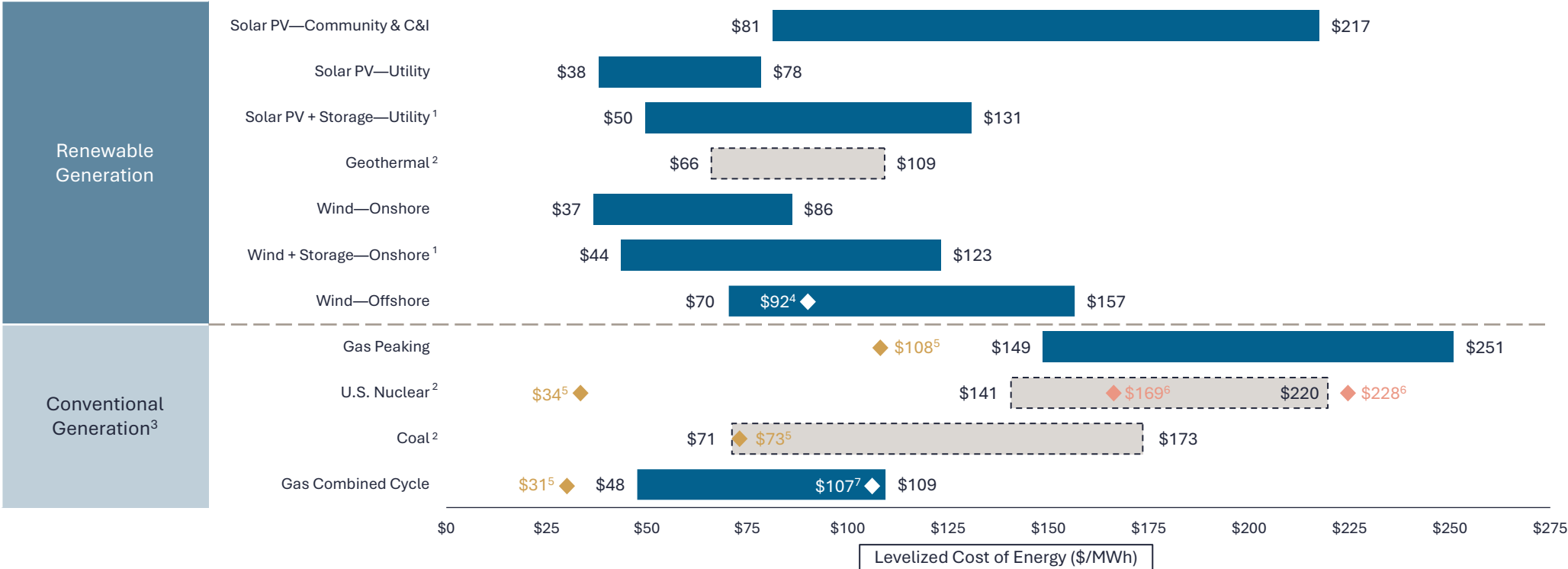
Lazard's Levelized Cost of Energy analysis addresses the following topics:

- Comparative LCOE analysis for various generation technologies on a \$/MWh basis, including sensitivities for U.S. federal tax subsidies, fuel prices, carbon pricing and cost of capital
- Illustration of how the LCOE of onshore wind, utility-scale solar and hybrid projects compare to the marginal cost of selected conventional generation technologies
- Historical LCOE comparison of various technologies
- Illustration of the historical LCOE declines for onshore wind and utility-scale solar
- Appendix materials, including:
 - An overview of the methodology utilized to prepare Lazard's LCOE analysis
 - A summary of the assumptions utilized in Lazard's LCOE analysis
 - Deconstruction of the LCOE for various generation technologies by capital cost, fixed operations and maintenance ("O&M") expense, variable O&M expense and fuel cost

Other factors would also have a potentially significant effect on the results contained herein but have not been examined in the scope of this current analysis. These additional factors, among others, may include: recent tariff-related cost impacts; implementation and interpretation of the full scope of the IRA; economic policy, transmission queue reform, network upgrades and other transmission matters, congestion, curtailment or other integration-related costs; permitting or other development costs, unless otherwise noted; and costs of complying with various environmental regulations (e.g., carbon emissions offsets or emissions control systems). This analysis is intended to represent a snapshot in time and utilizes a wide, but not exhaustive, sample set of Industry data. As such, we recognize and acknowledge the likelihood of results outside of our ranges. Therefore, this analysis is not a forecasting tool and should not be used as such given the complexities of our evolving Industry, grid and resource needs. Except as illustratively sensitized herein, this analysis does not consider the intermittent nature of selected renewables energy technologies or the related grid impacts of incremental renewable energy deployment. This analysis also does not address potential social and environmental externalities including, for example, the social costs and rate consequences for those who cannot afford distributed generation solutions as well as the long-term residual and societal consequences of various conventional generation technologies that are difficult to measure (e.g., airborne pollutants, greenhouse gases, etc.).

Levelized Cost of Energy Comparison—Version 18.0

Selected renewable energy generation technologies remain cost-competitive with conventional generation technologies under certain circumstances



Source: Lazard estimates and publicly available information.

Note: Here and throughout this analysis, unless otherwise indicated, the analysis assumes 60% debt at an 8% interest rate and 40% equity at a 12% cost. See page titled "Levelized Cost of Energy Comparison—Sensitivity to Cost of Capital" for cost of capital sensitivities.

1 Reflects the LCOE for a system composed of standalone generation plus standalone storage less the combined system-level synergies (assumed to be 10% of storage capital costs and 25% of inverter costs). The synergies capture potential cost reductions or efficiency gains from integrating generation and storage, such as shared interconnection infrastructure, improved energy dispatch, enhanced capacity utilization and operational efficiencies.

2 Given the limited public and/or observable data available for new-build geothermal, coal and nuclear projects, the LCOE presented herein reflects Lazard's LCOE v14.0 results adjusted for inflation and, for nuclear, are based on then-estimated costs of the Vogtle Plant. Coal LCOE does not include cost of transportation and storage.

3 The fuel cost assumptions for Lazard's LCOE analysis of gas-fired generation, coal-fired generation and nuclear generation resources are \$3.45/MMBTU, \$1.47/MMBTU and \$0.85/MMBTU, respectively, for year-over-year comparison purposes. See page titled "Levelized Cost of Energy Comparison—Sensitivity to Fuel Prices" for fuel price sensitivities.

4 Represents the illustrative midpoint LCOE for Dominion's Coastal Virginia Offshore Wind ("CVOW") project, based on the publicly disclosed capital cost of ~\$8.7 billion (excluding onshore transmission costs) and offshore wind estimates from Lazard. Dominion's projected LCOE for CVOW as of February 2025 is \$91/MWh in 2027 dollars, with an expected COD in 4Q 2026.

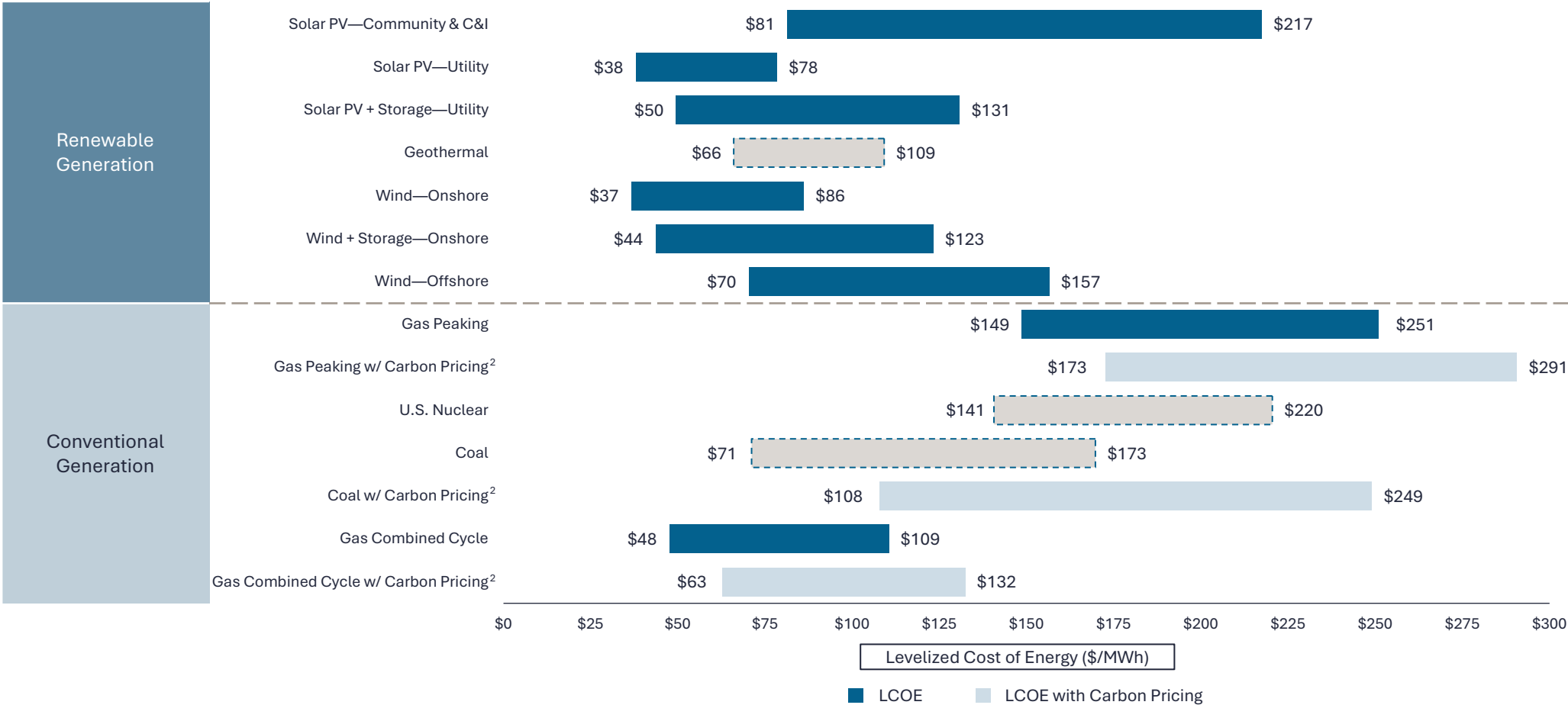
5 Reflects the average of the high and low LCOE marginal cost of operating fully depreciated gas peaking, gas combined cycle, coal and nuclear facilities, inclusive of decommissioning costs for nuclear facilities. Analysis assumes that the salvage value for a decommissioned gas or coal asset is equivalent to its decommissioning and site restoration costs. Inputs are derived from a benchmark of operating gas, coal and nuclear assets across the U.S. Capacity factors, fuel, variable and fixed operating expenses are based on upper- and lower-quartile estimates derived from Lazard's research. See page titled "Levelized Cost of Energy Comparison—New Build Renewable Generation vs. Marginal Cost of Conventional Generation" for additional details.

6 Represents illustrative LCOE values for Vogtle nuclear plant's units 3 and 4. The analysis is based on publicly available estimates and suggestions from selected industry experts, indicating a cost "learning curve" of ~30% between Vogtle units 3 and 4. Analysis assumes total operating capacity of ~2.2 GW, total capital cost of ~\$32.3 billion, capacity factor of ~97%, operating life of 70 years and other operating parameters estimated by Lazard's LCOE v14.0 results, adjusted for inflation.

7 Illustrative high case reflects elevated capital costs (\$2,400/kW – \$2,600/kW) based on recently observed market quotes for CCGT projects in early stages of development (post-2028 COD).

Levelized Cost of Energy Comparison—Sensitivity to Carbon Pricing

Carbon pricing is one avenue for policymakers to address carbon emissions; a carbon price range of \$40 – \$60/Ton¹ of carbon would increase the LCOE for certain conventional generation technologies, as indicated below



Source:

Lazard estimates and publicly available information.

Note:

Unless otherwise noted, the assumptions used in this sensitivity correspond to those used in the LCOE analysis as presented on the page titled “Levelized Cost of Energy Comparison—Version 18.0”. LCOE with Carbon Pricing is limited to carbon emissions directly related to generation and does not include the impacts of carbon pricing on embodied carbon.

1

The current administration no longer maintains an estimate of the monetized impacts of greenhouse gas emissions. Previous administrations estimated the social cost of carbon to range from \$5/Ton (first Trump Administration) to over \$200/Ton (Biden Administration).

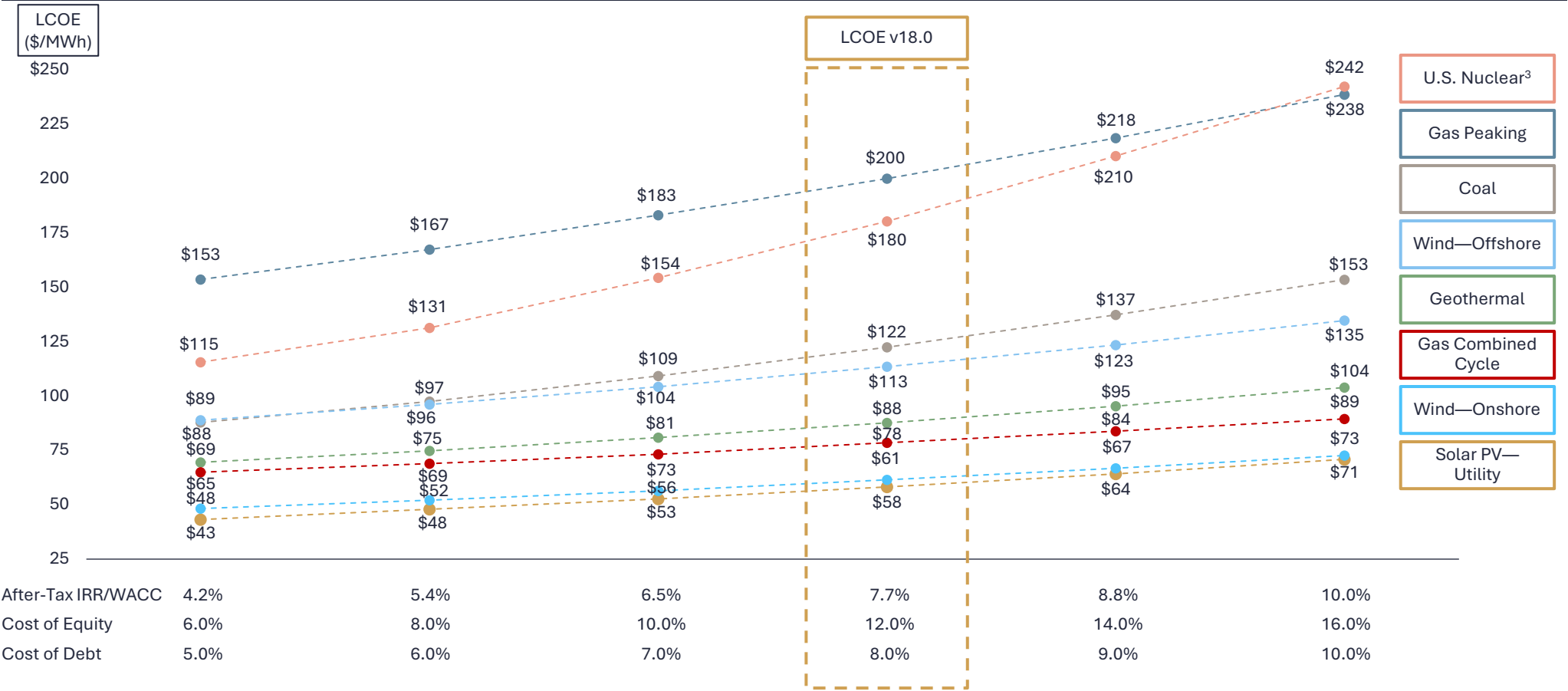
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The low and high ranges reflect the LCOE of selected conventional generation technologies including an illustrative carbon price of \$40/Ton and \$60/Ton, respectively.

Levelized Cost of Energy Comparison—Sensitivity to Cost of Capital¹

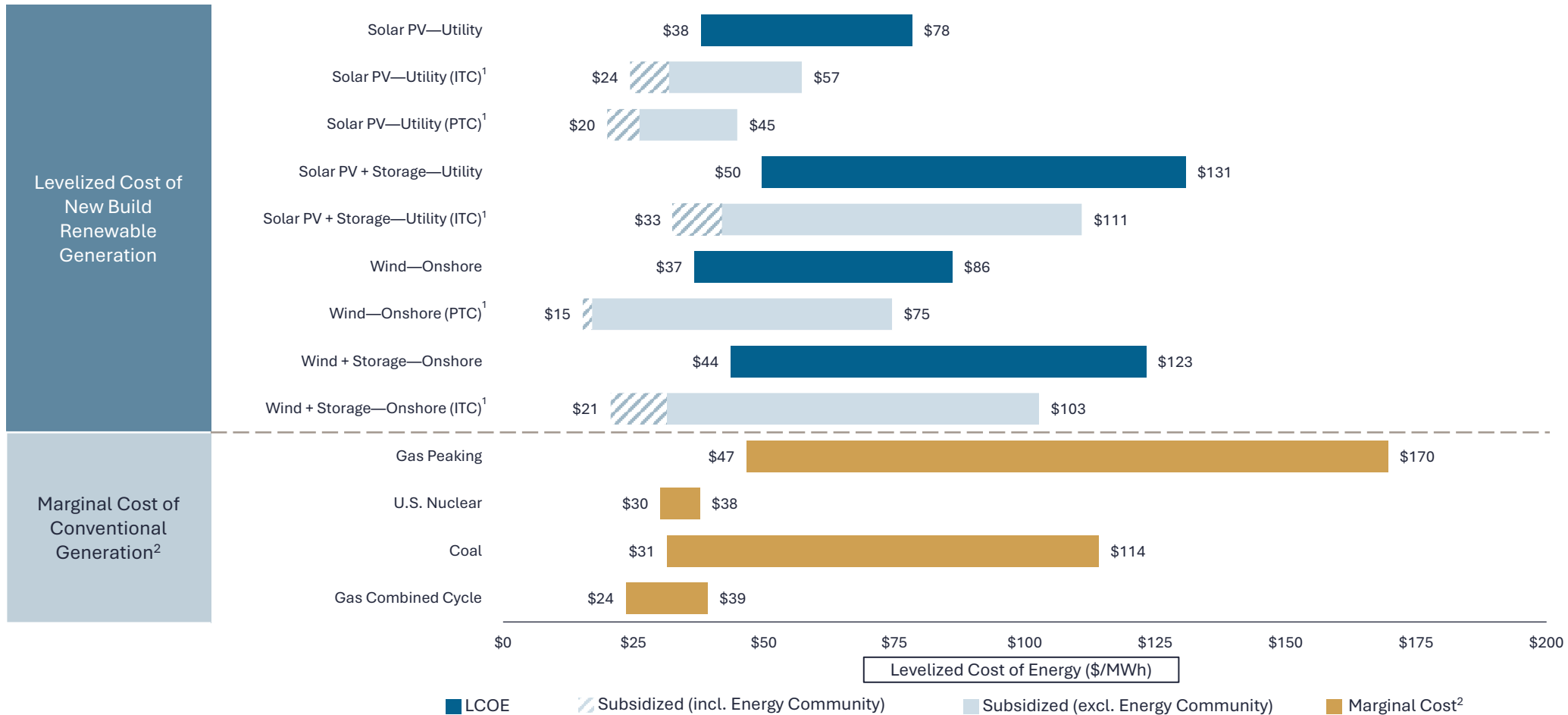
A key consideration in determining the LCOE for utility-scale generation technologies is the cost, and availability, of capital¹. In practice, this dynamic is particularly significant because the cost of capital for each asset is related to its specific operational characteristics and the resulting risk/return profile

Average LCOE²



Levelized Cost of Energy Comparison—New Build Renewable Generation vs. Marginal Cost of Conventional Generation

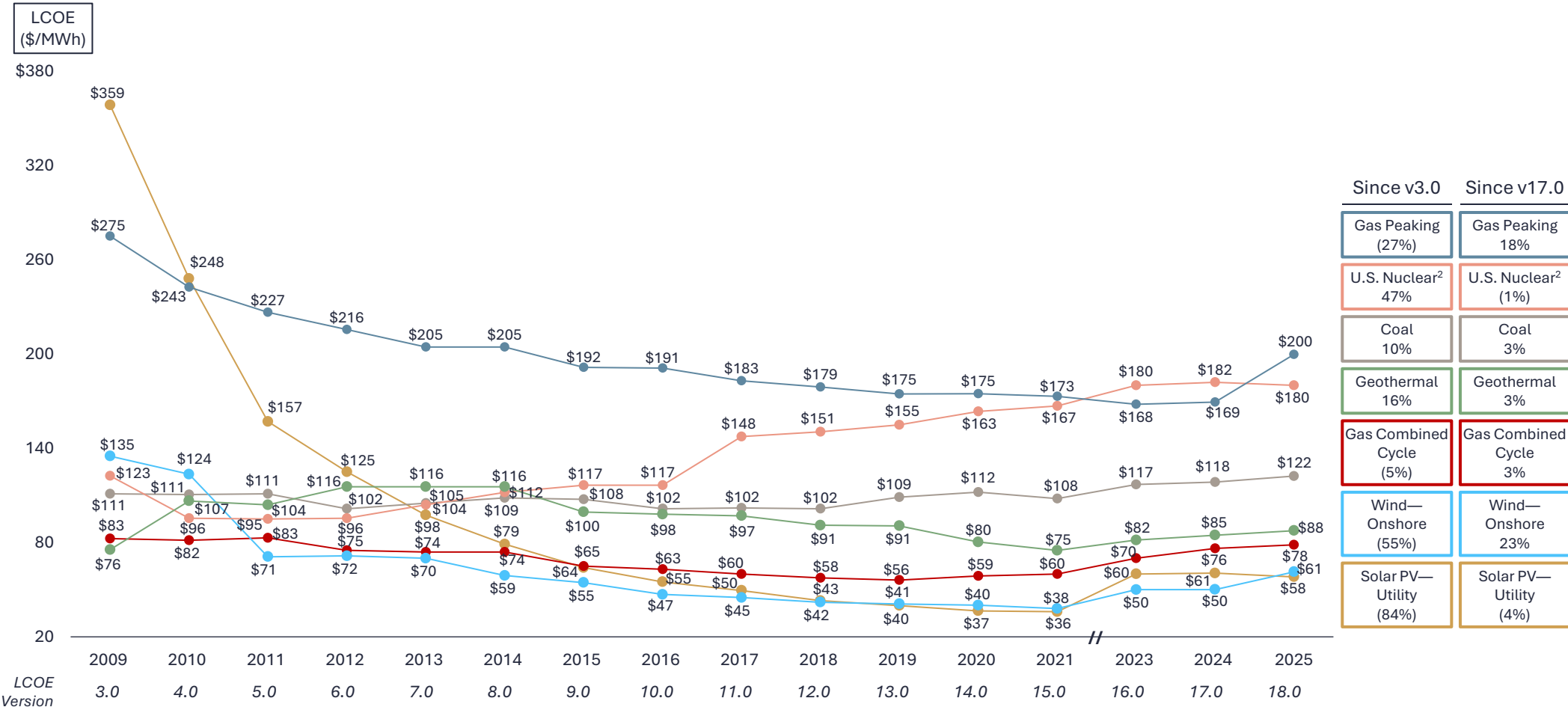
Certain renewable energy generation technologies have an LCOE that is competitive with the marginal cost of selected conventional generation technologies—notably, as incremental, intermittent renewable energy capacity is deployed and baseload gas-fired generation utilization rates increase, this gap closes, particularly in low gas pricing and high energy demand environments



Levelized Cost of Energy Comparison—Historical LCOE Comparison

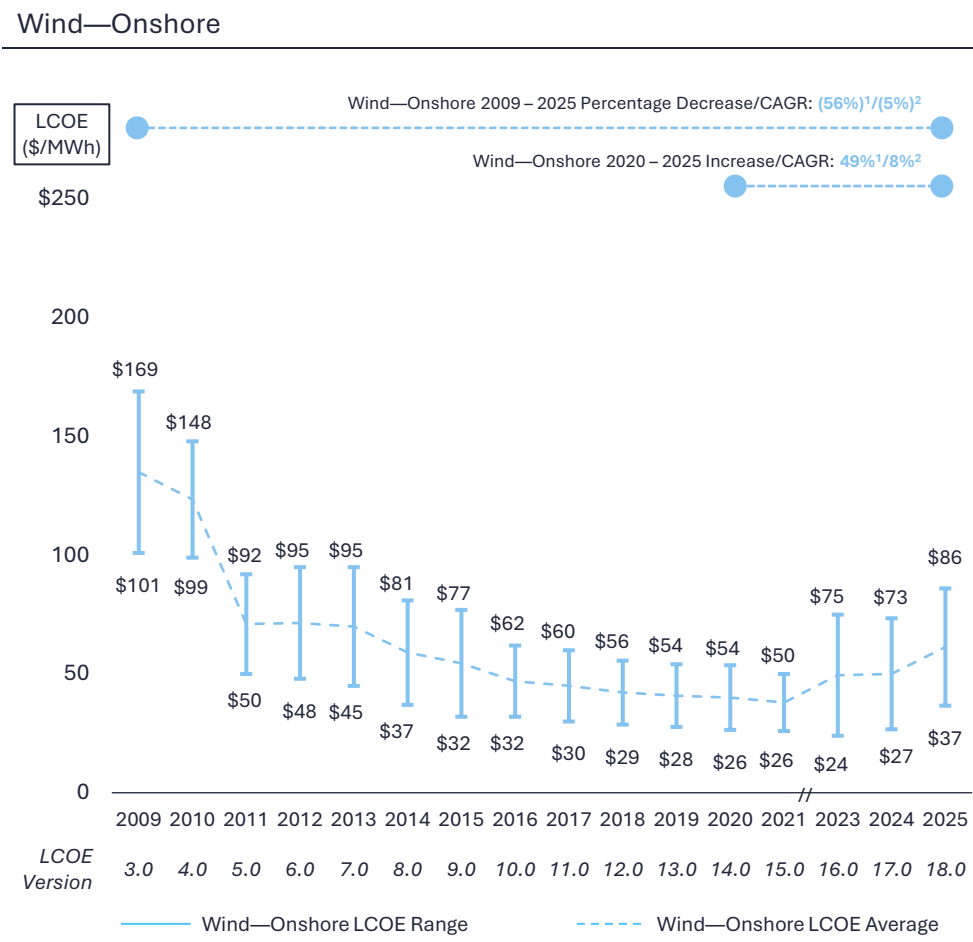
Lazard’s LCOE analysis indicates significant historical cost declines for utility-scale renewable energy generation technologies, which has begun to level out and even slightly increase in recent years

Selected Historical Average LCOE Values¹



Levelized Cost of Energy Comparison—Historical Renewable Energy LCOE

This year’s analysis shows a divergence in trends between wind and solar with solar costs declining slightly and wind costs increasing, likely reflecting the difference in supply chain conditions across each technology



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Energy Storage



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Lazard's Levelized Cost of Storage Analysis—Version 10.0

Introduction

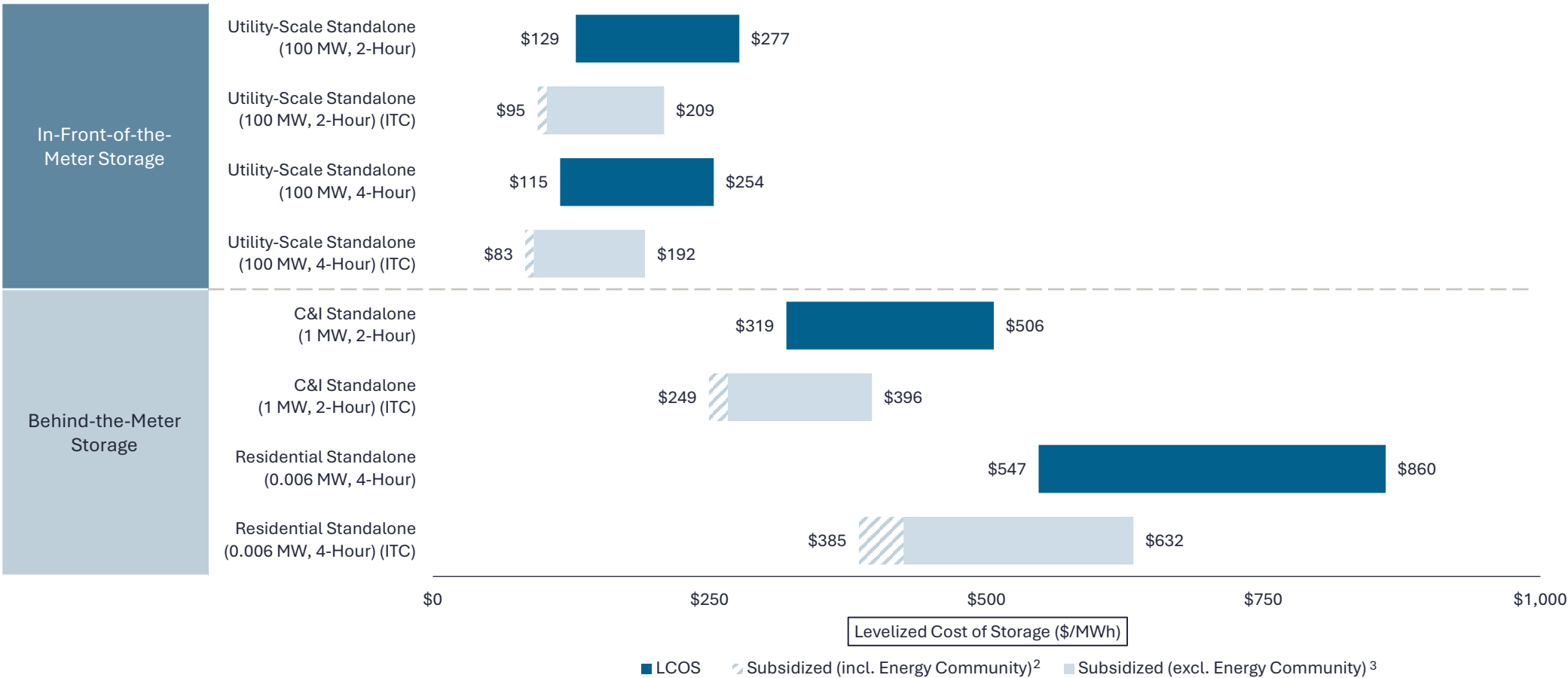
Lazard's Levelized Cost of Storage analysis addresses the following topics:

- LCOS Analysis:
 - Comparative LCOS analysis for various energy storage systems on a \$/MWh basis
 - Comparative LCOS analysis for various energy storage systems on a \$/kW-year basis
- Storage Value Snapshot Case Studies:
 - Overview of potential revenue applications for various energy storage systems
 - Overview of the Storage Value Snapshot Case Studies analysis and identification of selected geographies for each use case analyzed
 - Results from the Storage Value Snapshot Case Studies analysis
- Appendix Materials, including:
 - An overview of the use cases and operational parameters of selected energy storage systems for each use case analyzed
 - An overview of the methodology utilized to prepare Lazard's LCOS analysis
 - A summary of the assumptions utilized in Lazard's LCOS analysis
 - Deconstruction of the LCOS for various generation technologies by capital cost, fixed operations and maintenance ("O&M") expense and charging cost

Other factors would also have a potentially significant effect on the results contained herein but have not been examined in the scope of this current analysis. These additional factors, among others, may include: recent tariff-related cost impacts; implementation and interpretation of the full scope of the IRA; economic policy, transmission queue reform, network upgrades and other transmission matters; congestion, curtailment or other integration-related costs; permitting or other development costs, unless otherwise noted; and costs of complying with various regulations (e.g., federal import tariffs or labor requirements). This analysis also does not address potential social and environmental externalities as well as the long-term residual and societal consequences of various energy storage system technologies that are difficult to measure (e.g., resource extraction, end-of-life disposal, lithium-ion-related safety hazards, etc.). This analysis is intended to represent a snapshot in time and utilizes a wide, but not exhaustive, sample set of Industry data. As such, we recognize and acknowledge the likelihood of results outside of our ranges. Therefore, this analysis is not a forecasting tool and should not be used as such given the complexities of our evolving Industry, grid and resource needs.

Levelized Cost of Storage Comparison—Version 10.0 (\$/MWh)

Lazard’s LCOS analysis evaluates standalone energy storage systems on a levelized basis to derive cost metrics across energy storage use cases and configurations¹



Source: Lazard estimates and publicly available information.

Note: Here and throughout this section, unless otherwise indicated, the analysis assumes 20% debt at an 8% interest rate and 80% equity at a 12% cost, which is a different capital structure than Lazard’s LCOE analysis. Capital costs include the storage module, balance of system and power conversion equipment, collectively referred to as the energy storage system, equipment (where applicable) and EPC costs. Augmentation costs are not included in capital costs in this analysis and vary across use cases due to usage profiles and lifespans. Charging costs are assessed at the weighted average hourly pricing (wholesale energy prices) across an optimized annual charging profile of the asset. See Appendix B for charging cost assumptions and additional details. The projects are assumed to use a 5-year MACRS depreciation schedule.

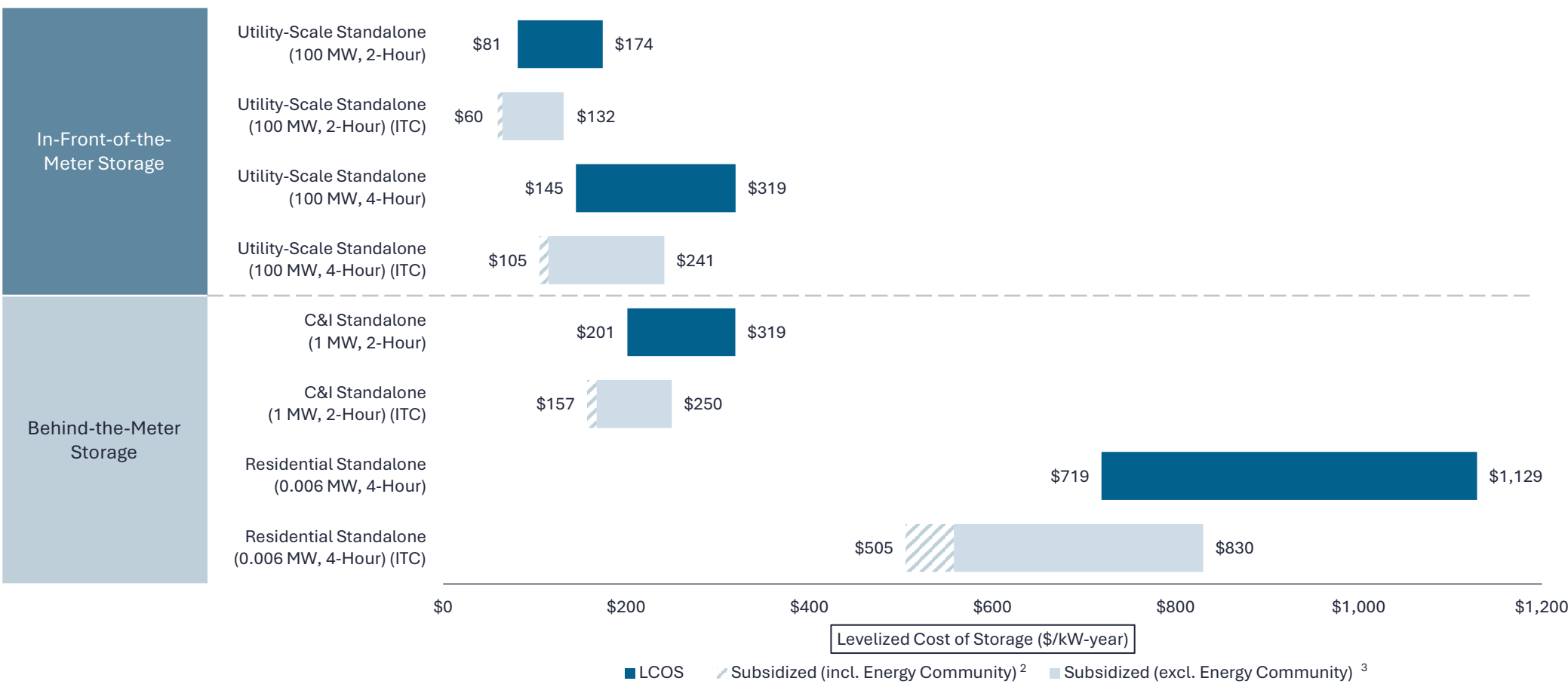
1 See Appendix B for a detailed overview of the use cases and operational parameters analyzed in the LCOS.

2 This sensitivity analysis assumes that projects qualify for the full ITC and have a capital structure that includes sponsor equity, debt and tax equity and also includes a 10% Energy Community adder.

3 This sensitivity analysis assumes that projects qualify for the full ITC and have a capital structure that includes sponsor equity, debt and tax equity.

Levelized Cost of Storage Comparison—Version 10.0 (\$/kW-year)

Lazard’s LCOS analysis evaluates standalone energy storage systems on a levelized basis to derive cost metrics across energy storage use cases and configurations¹



Levelized Cost of Storage Comparison—Historical LCOS Comparison

This year’s analysis shows notable declines in the LCOS of utility scale and C&I battery energy storage systems. Key drivers include both market dynamics—slower-than-expected EV demand and the resulting oversupply of cells—and technological advancements, including increased cell capacity and energy density



Storage Value Snapshot Case Studies—Revenue Potential for Selected Use Cases

The numerous potential sources of revenue available to energy storage systems reflect the benefits provided to customers and the grid

- The scope of revenue sources is limited to those captured by existing or soon-to-be commissioned projects—revenue sources that are not clearly identifiable or without publicly available data have not been analyzed

			Use Cases¹				
Description			Utility-Scale Standalone	Utility-Scale PV + Storage	Utility-Scale Wind + Storage	Commercial & Industrial Standalone	Commercial & Industrial PV + Storage
Wholesale	Demand Response—Wholesale	Manages high wholesale price or emergency conditions on the grid by calling on users to reduce or shift electricity demand				✓	✓
	Energy Arbitrage	Storage of inexpensive electricity to sell later at higher prices (only evaluated in the context of a wholesale market)	✓	✓	✓		
	Frequency Regulation	Provides immediate (4-second) power to maintain generation-load balance and prevent frequency fluctuations	✓	✓	✓		
	Resource Adequacy	Provides capacity to meet generation requirements at peak load	✓	✓	✓		
	Spinning/Non-Spinning Reserves	Maintains electricity output during unexpected contingency events (e.g., outages) immediately (spinning reserve) or within a short period of time (non-spinning reserve)	✓	✓	✓		
Utility	Demand Response—Utility	Manages high wholesale price or emergency conditions on the grid by calling on users to reduce or shift electricity demand				✓	✓
Customer	Bill Management	Allows reduction of demand charge using battery discharge and the daily storage of electricity for use when time of use rates are highest				✓	✓
	Incentives	Payments provided to residential and commercial customers to encourage the acquisition and installation of energy storage systems				✓	✓

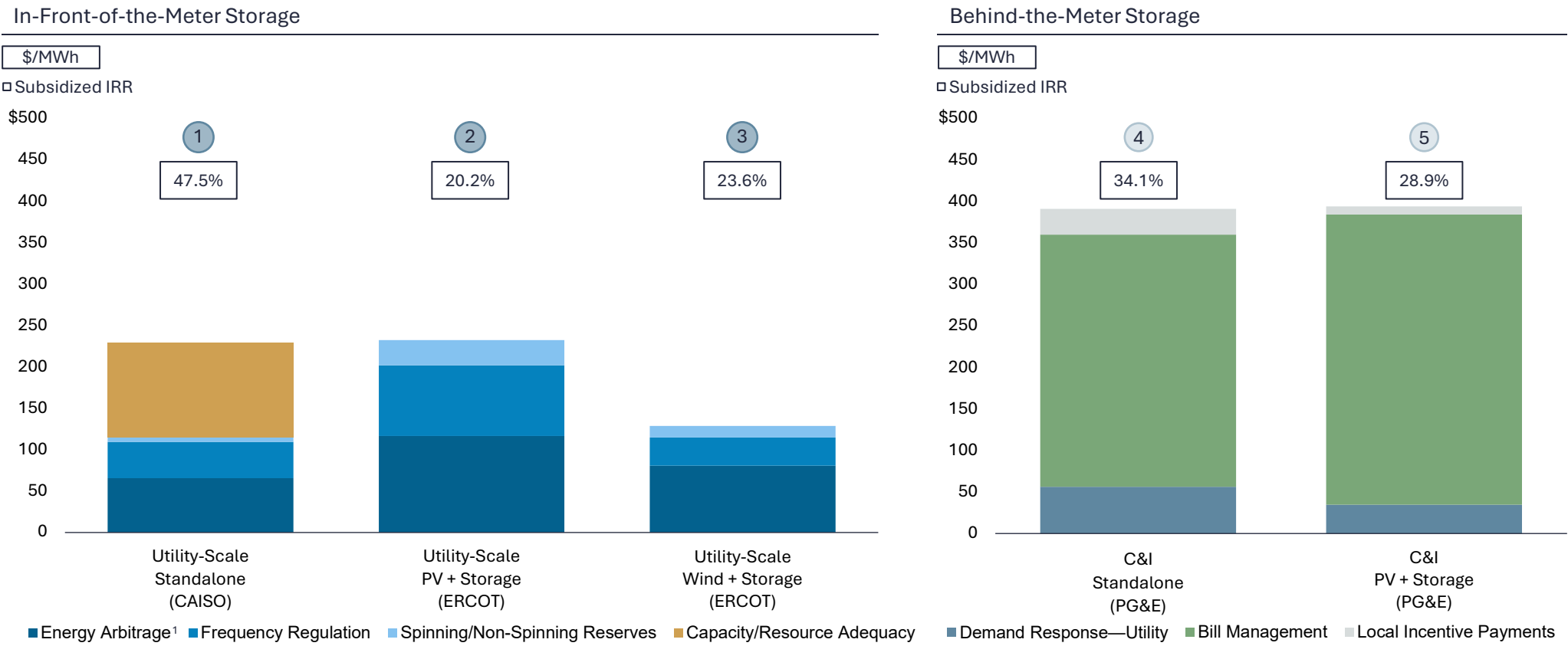
Storage Value Snapshot Case Studies—Overview

Lazard’s Storage Value Snapshots analyze the financial viability of illustrative energy storage systems designed for selected use cases and geographies

		Location	Description	Storage (MW)	Generation (MW)	Storage Duration (hours)	Revenue Streams
In-Front-of-the-Meter Storage	1	CAISO ¹ (SP-15)	Large-scale energy storage system	100	–	4	• Energy Arbitrage • Frequency Regulation • Resource Adequacy • Spinning/Non-Spinning Reserves
	2	ERCOT ² (South Texas)	Energy storage system designed to be paired with large solar PV facilities	50	100	4	
	3	ERCOT ² (South Texas)	Energy storage system designed to be paired with large wind generation facilities	50	100	4	
Behind-the-Meter Storage	4	PG&E ³ (California)	Energy storage system designed for behind-the-meter peak shaving and demand charge reduction for C&I energy users	1	–	2	• Demand Response—Utility • Bill Management • Incentives • Tariff Settlement, Demand Response Participation, Avoided Costs to Commercial Customer and Local Capacity Resource Programs
	5	PG&E ³ (California)	Energy storage system designed for behind-the-meter peak shaving and demand charge reduction services for C&I energy users	0.5	1	4	

Storage Value Snapshot Case Studies—Results

Project economics evaluated in the Storage Value Snapshot Case Studies continue to evolve year-over-year as costs change and the value of revenue streams adjust to reflect underlying market conditions, utility rate structures and policy developments. Notably, this year capacity/resource adequacy payments nearly doubled which, combined with LCOS declines, significantly increased project returns



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Energy System

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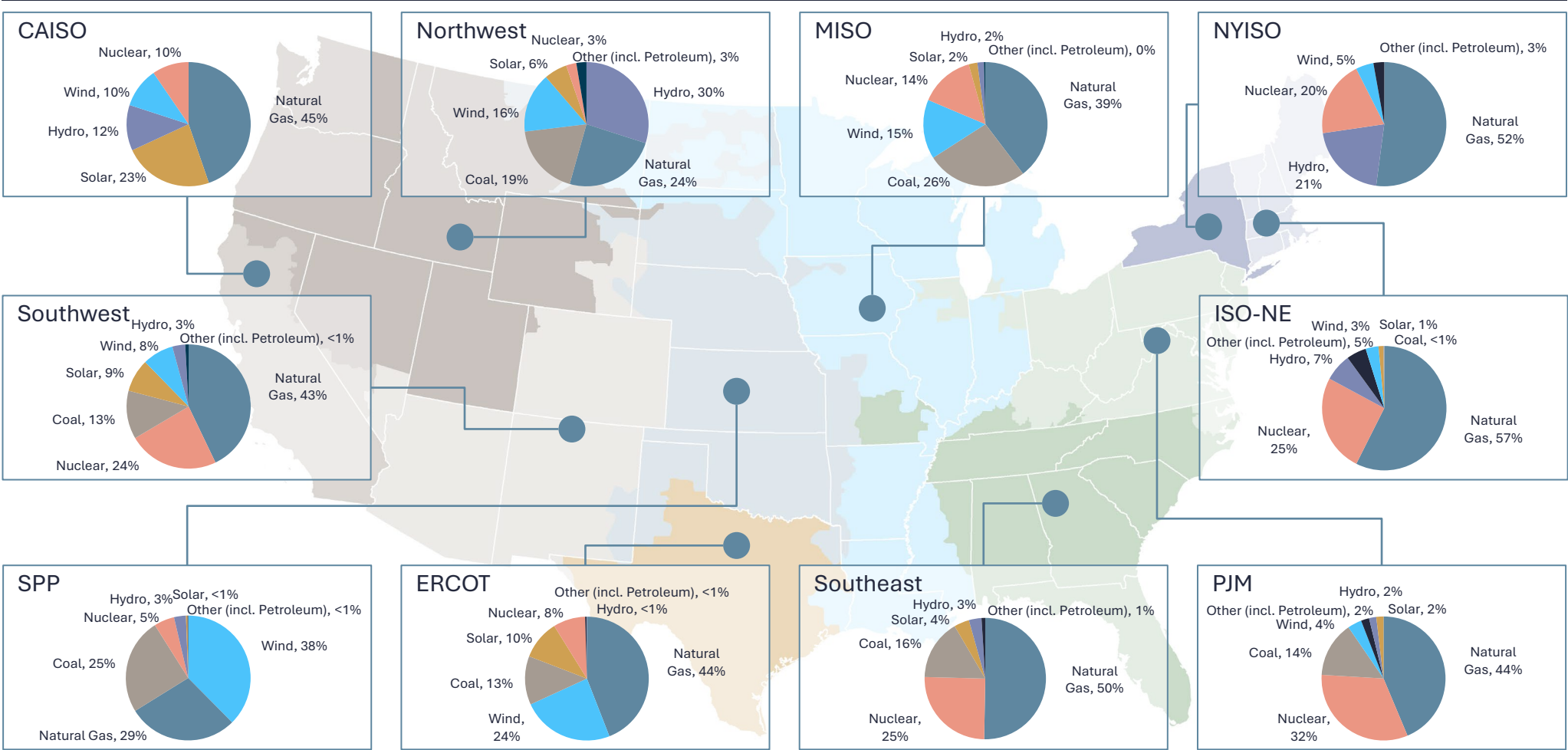
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Cost of Firming Intermittency

Market Overview—Current Generation Mix

The current generation mix across the U.S. varies significantly by market—resource availability, operational constraints, load profiles, transmission infrastructure, seasonal weather patterns and regulatory constructs, among other factors, are key drivers of such variation

2024 Generation Mix by Region



Market Overview—Current Firming Cost Frameworks

Many grid operators and utilities use effective load-carrying capability (“ELCC”) to measure the reliability of new power generation resources to contribute to the electricity grid at key periods of demand, particularly intermittent ones like wind and solar. Combined with the net cost of new entry (“Net CONE”)¹, as determined by the grid operator, ELCC helps to guide decisions on resource planning, capacity adequacy and system reliability. Balancing authorities (“BA”s) such as MISO, CAISO, SPP, PJM and ERCOT have adopted ELCC accreditation frameworks to ensure a reliable and efficient grid

- ELCC measures the performance of a resource at times of greatest “capacity need” for the system, where capacity need is a function of electricity demand patterns and the generation mix in each region—in general, the higher the renewable resource penetration, the lower the ELCC accreditation for each additional renewable resource

	BA-Specified “Firming” Source	ELCC Values ²	Net CONE ¹ (\$/kW-month)	Selected Market Commentary
MISO	Natural Gas Peaker	Solar: 39% Wind: 26%	\$10.03	• In March 2024, MISO adopted the FERC Reliability Availability and Need (“RAN”) seasonal capacity construct for wind and solar resources • Seasonal wind accredited capacity values are 18.1% for summer, 18.6% for fall, 53.1% for winter and 18.0% for spring • Solar capacity values are 50% for all seasons except winter, which is 5%
CAISO	4-Hour Lithium-Ion Battery	Solar: 7% PV + Storage ³ : 41% Wind: 12%	\$18.92	• Increasing levels of solar penetration in CAISO have shifted peak demand later in the day, reducing the ELCC value for solar • CAISO significantly reduced ELCC values for 4-hour battery storage systems, driven by significant growth in 4-hour storage capacity
SPP	Natural Gas Peaker	Solar: 51% Wind: 20%	\$8.38	• SPP published seasonal accreditation values based on 2024, assigning separate values to resources for summer and winter seasons • Summer wind and solar contributions are 15.2% and 25.5%, respectively, whereas winter values shift to 39.1% for wind and 62.2% for solar
PJM	Natural Gas Peaker	Solar: 12% PV + Storage ³ : 33% Wind: 38%	\$10.29	• PJM adopted a new, marginal ELCC methodology to begin in the 2025/2026 delivery year that reduces the reliability value of highly correlated resources, such as solar and short-duration storage⁴ • The update is expected to better capture expected resource performance during system peak
ERCOT	Natural Gas Peaker	Solar: 38% Wind: 25%	\$9.92	• ERCOT maintains notably high ELCC values despite having the highest renewable penetration by capacity of the U.S. regulatory markets • ERCOT updates its capacity scheme every three years; the most recent publication was December 2022

Source: Publicly available information.

1 Net “CONE” is defined as capital and operating costs less expected market revenues for a new, firm resource (e.g., gas peaker or battery storage). Net CONE is established by the respective balancing authority.

2 ELCC values are calculated by the respective balancing authority. ELCC is an indicator of the incremental reliability contribution of a given resource to the electricity grid based on its contribution to meeting peak electricity demand. For example, a 1 MW wind resource with a 15% ELCC provides 0.15 MW of capacity contribution and would need to be supplemented by 0.85 MW of additional firm capacity to represent the addition of 1 MW of firm system capacity. Where seasonal accreditation values exist, values have been annualized.

3 For PV + Storage cases, the effective ELCC value is represented. CAISO and PJM assess ELCC values separately for the PV and storage components of a system. Storage ELCC value is provided only for the capacity that can be charged directly by the accompanying resource up to the energy required for a 4-hour discharge during peak load. Any capacity available in excess of the 4-hour maximum discharge is attributed to the system at the solar ELCC. ELCC values for storage range from 55% to 75% for PJM and CAISO, respectively.

4 This year’s analysis does not reflect this future methodology.

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Cost of Firming Intermittency—Methodology

Lazard's Cost of Firming Intermittency analysis builds on the LCOE results by evaluating system-level costs associated with supplementing intermittent renewable energy on the grid with firm capacity to ensure reliable electricity delivery during peak demand periods. The analysis utilizes ELCC and Net CONE values assessed and published by grid operators for each regional market to determine these costs

- The firm capacity value of a new resource is calculated as $\text{Nameplate Capacity} \times \text{ELCC \%}$, where:
 - Nameplate Capacity of a resource refers to its maximum potential energy output, and
 - ELCC measures the performance of a resource at times of greatest “capacity need” for the system, where capacity need is a function of electricity demand patterns and the generation mix in each region
- Over time, increased renewable penetration or changes in demand patterns can shift the timing of the capacity need, impacting ELCC
- The remaining non-firm capacity ($\text{Nameplate Capacity} \times (1 - (\text{ELCC \%}))$) is “firmed” at the Net CONE, a \$/kW-month figure which is intended to reflect capital and operating costs less expected market revenues for a new, firm resource (e.g., gas peaker or battery storage)
 - Net CONE is assessed and published by grid operators for each regional market

In the following analysis, the Levelized Firming Cost is defined as the additional capacity payment, priced at Net CONE, required to bring the ELCC of the combined system (intermittent and firming resource) to 100%. The LCOE plus Levelized Firming Cost varies between ISOs, due to (1) the standalone LCOE in the region based on regional capacity factor for wind or solar, (2) the ELCC value of the standalone renewable resource and (3) the region's Net CONE

$$\frac{\text{Nameplate Capacity (kW)} \times (1 - \text{ELCC (\%)}) \times \text{Net CONE (\$/kW-month)} \times 12 \text{ Months}}{\text{Nameplate Capacity (MW)} \times \text{Regional Capacity Factor (\%)} \times 8,760 \text{ Hours}}$$

**Levelized Firming Cost
(\$/MWh)**



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Levelized Cost of Energy Comparison—Methodology

(\$ in millions, unless otherwise noted)

Lazard's LCOE analysis consists of creating a power plant model representing an illustrative project for each relevant technology and solving for the \$/MWh value that results in a levered IRR equal to the assumed cost of equity (see subsequent "Key Assumptions" pages for detailed assumptions by technology)

		Unsubsidized Onshore Wind — Low Case Sample Illustrative Calculations							Key Assumptions ⁵	
Year ¹		0	1	2	3	4	5	30		
Capacity (MW)	(A)		300	300	300	300	300	300	Capacity (MW)	300
Capacity Factor	(B)		55%	55%	55%	55%	55%	55%	Capacity Factor	55%
Total Generation ('000 MWh)	(C)* = (A) x (B)		1,445	1,445	1,445	1,445	1,445	1,445	Fuel Cost (\$/MMBtu)	\$0.00
Levelized Energy Cost (\$/MWh)	(D)		\$36.7	\$36.7	\$36.7	\$36.7	\$36.7	\$36.7	Heat Rate (Btu/kWh)	0
Total Revenues	(E)* = (C) x (D)		\$53.0	\$53.0	\$53.0	\$53.0	\$53.0	\$53.0	Fixed O&M (\$/kW-year)	\$24.5
Total Fuel Cost	(F)		--	--	--	--	--	--	Variable O&M (\$/MWh)	\$0.0
Total O&M	(G)*		7.4	7.5	7.7	7.9	8.0	14.0	O&M Escalation Rate	2.25%
Total Operating Costs	(H) = (F) + (G)		\$7.4	\$7.5	\$7.7	\$7.9	\$8.0	\$14.0	Capital Structure	
EBITDA	(I) = (E) - (H)		\$45.7	\$45.5	\$45.3	\$45.1	\$45.0	\$39.0	Debt	60.0%
Debt Outstanding - Beginning of Period	(J)		\$342.0 ²	\$339.0	\$335.7	\$332.2	\$328.4	\$28.1	Cost of Debt	8.0%
Debt - Interest Expense	(K)		(27.4)	(27.1)	(26.9)	(26.6)	(26.3)	(2.3)	Equity	40.0%
Debt - Principal Payment	(L)		(3.0)	(3.3)	(3.5)	(3.8)	(4.1)	(28.1)	Cost of Equity	12.0%
Levelized Debt Service	(M) = (K) + (L)		(\$30.4)	(\$30.4)	(\$30.4)	(\$30.4)	(\$30.4)	(\$30.4)	Taxes and Tax Incentives:	
EBITDA	(I)		\$45.7	\$45.5	\$45.3	\$45.1	\$45.0	\$39.0	Combined Tax Rate	40%
Depreciation (MACRS)	(N)		(114.0)	(182.4)	(109.4)	(65.7)	(65.7)	0.0	Economic Life (years) ⁶	30
Interest Expense	(K)		(27.4)	(27.1)	(26.9)	(26.6)	(26.3)	39.0	MACRS Depreciation (Year Schedule)	5
Taxable Income	(O) = (I) + (N) + (K)		(\$95.7)	(\$164.0)	(\$91.0)	(\$47.1)	(\$47.0)	(\$2.3)	Capex	
Tax Benefit (Liability)³	(P) = (O) x (tax rate)		\$38.5	\$65.9	\$36.6	\$18.9	\$18.9	(\$14.8)	EPC Costs (\$/kW)	\$1,900
After-Tax Net Equity Cash Flow	(Q) = (I) + (M) + (P)	(\$228.0)⁴	\$53.7	\$81.0	\$51.5	\$33.7	\$33.5	(\$6.2)	Additional Owner's Costs (\$/kW)	\$0
IRR For Equity Investors			12%						Transmission Costs (\$/kW)	\$0
									Total Capital Costs (\$/kW)	\$1,900
									Total Capex (\$m)	\$570

Source: Lazard estimates and publicly available information.

Note: Numbers presented for illustrative purposes only.

* Denotes unit conversion.

1 Assumes half-year convention for discounting purposes.

2 Reflects initial debt financing to fund capex.

3 Assumes full monetization of tax benefits or losses immediately.

4 Reflects initial cash outflow from equity investors to fund capex.

5 Reflects a "key" subset of all assumptions for methodology illustration purposes only. Does not reflect all assumptions.

6 Economic life sets debt amortization schedule.

■ Technology-Dependent

■ Consistent Across
Versions/Technologies

Levelized Cost of Energy—Key Assumptions

		Renewable Energy: Solar PV			
	Units	Community and C&I		Utility	
		Low	High	Low	High
Net Facility Output	MW	2.0		150	
Total Capital Cost	\$/kW	\$1,600	– \$3,300	\$1,150	– \$1,600
Fixed O&M	\$/kW-yr	\$13.00	– \$20.00	\$11.00	– \$14.00
Variable O&M	\$/MWh	—		—	
Heat Rate	Btu/kWh	—		—	
Capacity Factor	%	20%	– 15%	30%	– 20%
Fuel Price	\$/MMBTU	—		—	
Construction Time	Months	6		15	
Facility Life	Years	30		35	
Levelized Cost of Energy	\$/MWh	\$81	– \$217	\$38	– \$78

Levelized Cost of Energy—Key Assumptions (cont'd)

		Renewable Energy					
	Units	Geothermal		Wind—Onshore		Wind—Offshore	
		Low	High	Low	High	Low	High
Net Facility Output	MW	250		300		900	
Total Capital Cost	\$/kW	\$5,000	– \$6,460	\$1,900	– \$2,300	\$3,450	– \$6,550
Fixed O&M	\$/kW-yr	\$14.50	– \$15.75	\$24.50	– \$40.00	\$60.00	– \$91.50
Variable O&M	\$/MWh	\$9.05	– \$24.80	—		—	
Heat Rate	Btu/kWh	—		—		—	
Capacity Factor	%	90%	– 80%	55%	– 30%	55%	– 45%
Fuel Price	\$/MMBTU	—		—		—	
Construction Time	Months	36		18		24	
Facility Life	Years	25		30		30	
Levelized Cost of Energy	\$/MWh	\$66	– \$109	\$37	– \$86	\$70	– \$157

Levelized Cost of Energy—Key Assumptions (cont'd)

		Renewable Energy: Hybrid Generation + Storage					
		Solar PV + Storage—Utility			Wind + Storage—Onshore		
Units		Low		High	Low		High
Storage							
Power Rating	MW	50			50		
Duration	Hours	4			4		
Usable Energy	MWh	200			200		
90% Depth of Discharge Cycles/Year	%	350			350		
Roundtrip Efficiency	%	92%			92%		
Inverter Cost	\$/kW	\$19	–	\$50	\$19	–	\$50
Total Capital Cost (excl. Inverter)	\$/kWh	\$122	–	\$313	\$122	–	\$313
Storage O&M	\$/kWh	\$3.00	–	\$8.02	\$3.00	–	\$8.02
Generation							
Capacity	MW	100			100		
Capacity Factor	%	30.0%	–	20.0%	55.0%	–	30.0%
Project Life	Years	35			30		
Total Capital Cost	\$/kW	\$1,150	–	\$1,600	\$1,900	–	\$2,300
Fixed O&M	\$/kW	\$11.00	–	\$14.00	\$24.50	–	\$40.00
Extended Warranty Start	Year	3			3		
Warranty Expense % of Capital Costs	%	0.7%	–	1.9%	0.7%	–	1.9%
Charging Cost	\$/MWh	\$0.00			\$0.00		
Unsubsidized LCOE	\$/MWh	\$50	–	\$131	\$44	–	\$123

Levelized Cost of Energy—Key Assumptions (cont'd)

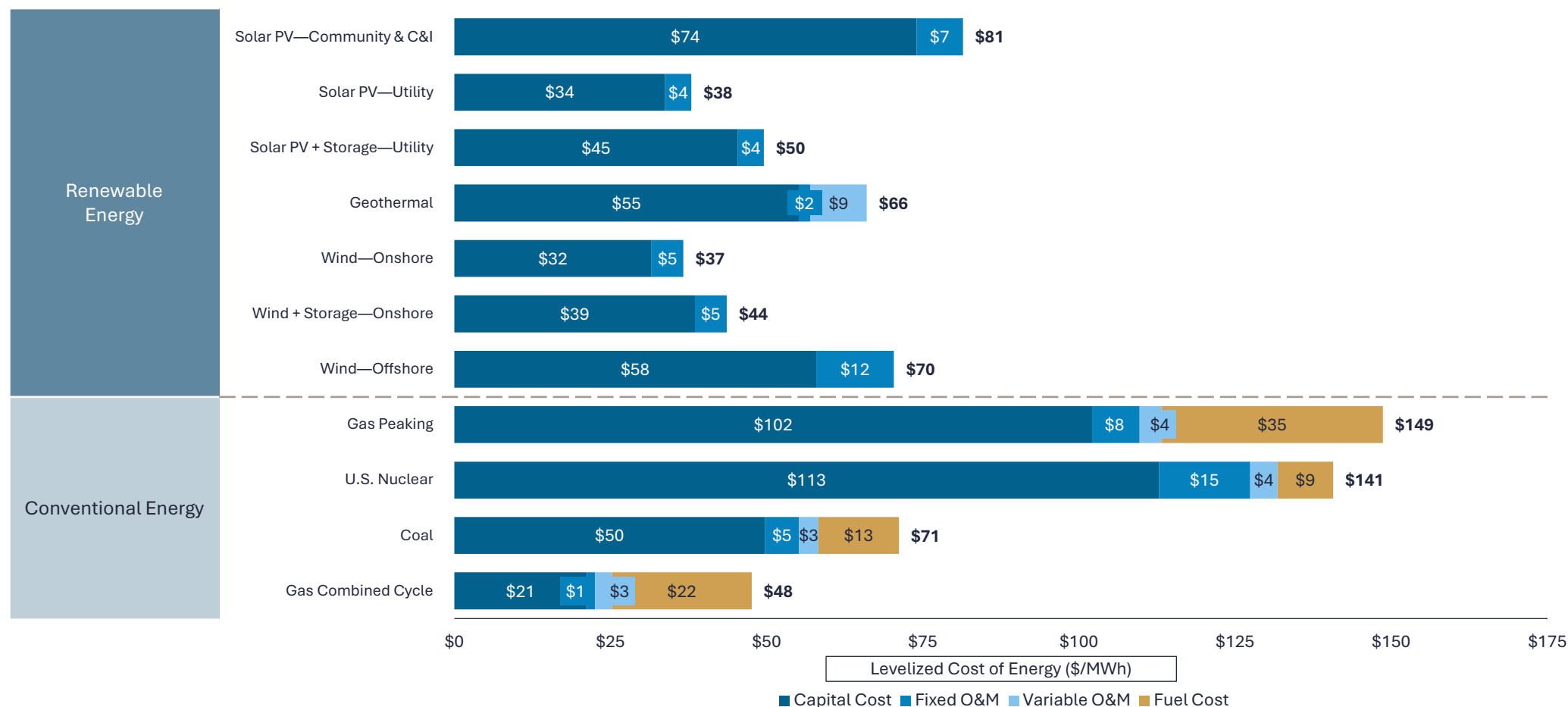
	Units	Conventional Energy							
		Gas Peaking (New Build)		U.S. Nuclear (New Build)		Coal (New Build)		Gas Combined Cycle (New Build)	
		Low	High	Low	High	Low	High	Low	High
Net Facility Output	MW	550	– 175	2,200		600		1,225	– 750
Total Capital Cost	\$/kW	\$1,150	– \$1,450	\$9,020 – \$14,820		\$3,405 – \$7,210		\$1,200	– \$1,600
Fixed O&M	\$/kW-yr	\$10.00	– \$17.00	\$136.00 – \$158.00		\$40.85 – \$94.35		\$10.00	– \$25.50
Variable O&M	\$/MWh	\$3.50	– \$5.00	\$4.40 – \$5.15		\$3.10 – \$5.70		\$2.75	– \$5.00
Heat Rate	Btu/kWh	10,275	– 11,175	10,450		8,750 – 12,000		6,475	– 6,550
Capacity Factor	%	15%	– 10%	92% – 89%		85% – 65%		90%	– 30%
Fuel Price	\$/MMBTU	\$3.45		\$0.85		\$1.47		\$3.45	
Construction Time	Months	24		84		60 – 66		24	
Facility Life	Years	30		70		40		30	
Levelized Cost of Energy	\$/MWh	\$149	– \$251	\$141 – \$220		\$71 – \$173		\$48	– \$109

Levelized Cost of Energy—Key Assumptions (cont'd)

		Marginal Cost of Selected Existing Conventional Generation											
	Units	Gas Peaking (Operating)			U.S. Nuclear (Operating)			Coal (Operating)			Gas Combined Cycle (Operating)		
		Low		High	Low		High	Low		High	Low		High
Net Facility Output	MW	240	–	50	2,200			600			550		
Total Capital Cost	\$/kW	\$0			\$0			\$0			\$0		
Fixed O&M	\$/kW-yr	\$4.00	–	\$6.10	\$89.00	–	\$121.60	\$21.70	–	\$33.80	\$8.90	–	\$13.60
Variable O&M	\$/MWh	\$2.70	–	\$9.30	\$2.70	–	\$3.90	\$3.20	–	\$7.20	\$0.80	–	\$1.80
Heat Rate	Btu/kWh	10,900	–	12,550	10,400	–	10,400	10,250	–	11,800	6,950	–	7,475
Capacity Factor	%	5%	–	1%	91%	–	87%	49%	–	7%	62%	–	17%
Fuel Price	\$/MMBtu	\$2.50	–	\$2.90	\$0.80	–	\$0.80	\$1.70	–	\$2.40	\$2.50	–	\$2.90
Construction Time	Months	24			84			60			24		
Facility Life	Years	30			70			40			30		
Levelized Cost of Energy	\$/MWh	\$47	–	\$170	\$30	–	\$38	\$31	–	\$114	\$24	–	\$39

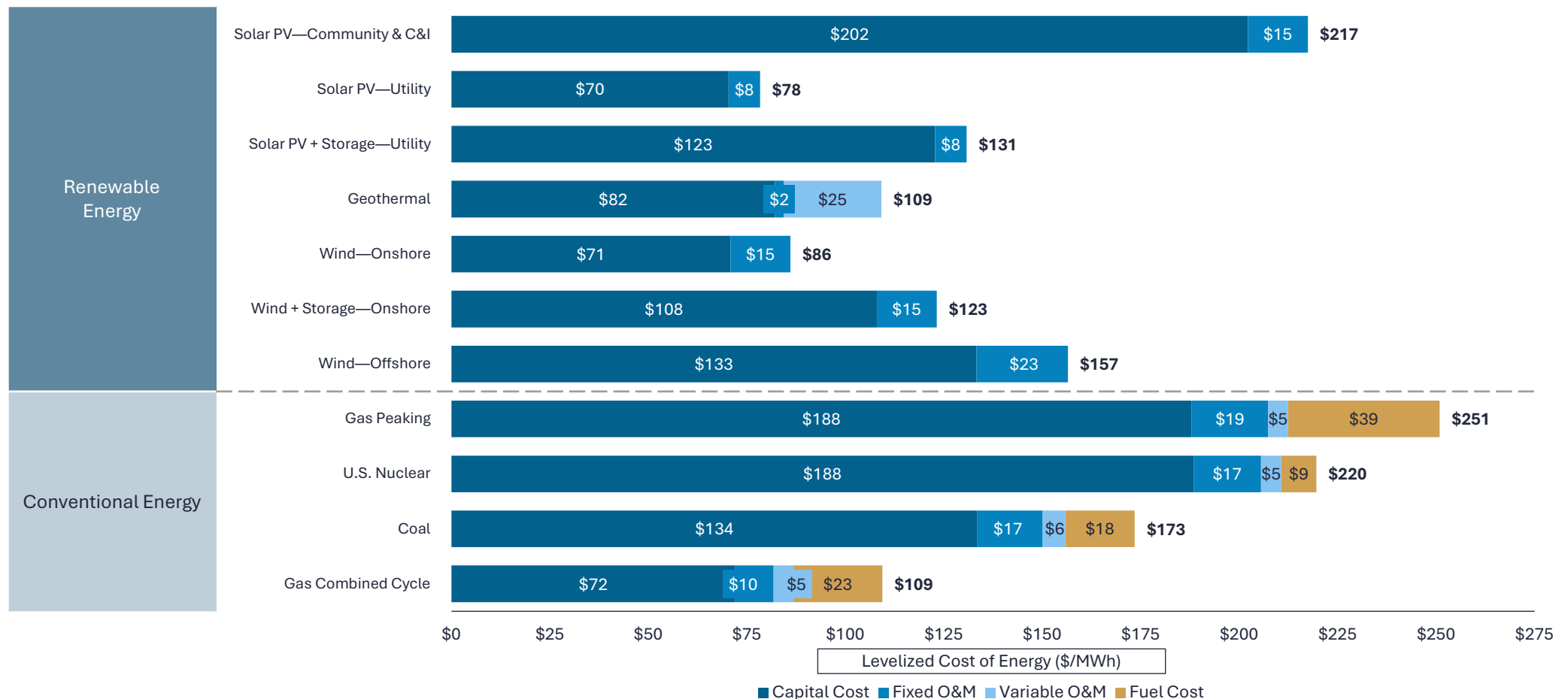
Levelized Cost of Energy Components—Low End (\$/MWh)

Certain renewable energy generation technologies are already cost-competitive with conventional generation technologies; key factors regarding the continued cost decline of renewable energy generation technologies are the ability of technological development and Industry scale to continue lowering operating expenses and capital costs for renewable energy generation technologies



Levelized Cost of Energy Components—High End (\$/MWh)

Certain renewable energy generation technologies are already cost-competitive with conventional generation technologies; key factors regarding the continued cost decline of renewable energy generation technologies are the ability of technological development and Industry scale to continue lowering operating expenses and capital costs for renewable energy generation technologies





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LCOS v10.0

Levelized Cost of Storage Comparison—Methodology

(\$ in millions, unless otherwise noted)

Lazard's LCOS analysis consists of creating a power plant model representing an illustrative project for each relevant technology and solving for the \$/MWh value that results in a levered IRR equal to the assumed cost of equity (see subsequent "Key Assumptions" page for detailed assumptions by technology)

Subsidized Utility-Scale Standalone (100 MW/200 MWh)—Low Case Sample Calculations

Year ¹		0	1	2	3	4	5	20
Capacity (MW)	(A)		100	100	100	100	100	100
Available Capacity (MW)		110	109	107	104	102	110	110
Total Generation ('000 MWh) ²	(B)*		63	63	63	63	63	63
Levelized Storage Cost (\$/MWh)	(C)		\$95	\$95	\$95	\$95	\$95	\$95
Total Revenues	(D)* = (B) x (C)		\$6.0	\$6.0	\$6.0	\$6.0	\$6.0	\$6.0
Total Charging Cost ³	(E)		(2.3)	(2.3)	(2.4)	(2.4)	(2.5)	(3.3)
Total O&M, Warranty, & Augmentation ⁴	(F)*		(0.6)	(0.6)	(0.8)	(0.8)	(2.6)	(1.1)
Total Operating Costs	(G) = (E) + (F)		(\$2.9)	(\$2.9)	(\$3.2)	(\$3.3)	(\$5.1)	(\$4.5)
EBITDA	(H) = (D) - (G)		\$3.1	\$3.0	\$2.8	\$2.7	\$0.9	\$1.5
Debt Outstanding - Beginning of Period	(I)		\$6.8 ⁵	\$6.6	\$6.4	\$6.3	\$6.1	\$0.6
Debt - Interest Expense	(J)		(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	(0.1)
Debt - Principal Payment	(K)		(0.1)	(0.2)	(0.2)	(0.2)	(0.2)	(0.6)
Levelized Debt Service	(L) = (J) + (K)		(0.7)	(0.7)	(0.7)	(0.7)	(0.7)	(0.7)
EBITDA	(H)		\$3.1	\$3.0	\$2.8	\$2.7	\$0.9	\$1.5
Depreciation (MACRS)	(M)		(5.4)	(8.6)	(5.2)	(3.1)	(3.1)	0.0
Interest Expense	(J)		(0.5)	1.7	0.0	0.0	0.0	(0.5)
Taxable Income	(N) = (H) + (M) + (J)		(\$2.9)	(\$3.9)	(\$2.4)	(\$0.4)	(\$2.2)	\$1.1
Tax Benefit (Liability)	(O) = (N) x (Tax Rate)		\$1.2	\$1.6	\$1.0	\$0.2	\$0.9	(\$0.4)
Federal Investment Tax Credit (ITC)	(P)		\$13.5	\$0.0	\$0.0	\$0.0	\$0.0	\$0.0
After-Tax Net Equity Cash Flow	(Q) = (H) + (L) + (O) + (P)		(\$27.0)⁶	\$17.0	\$3.9	\$3.1	\$2.2	\$1.1
IRR For Equity Investors			12.0%					

Key Assumptions ⁷	
Power Rating (MW)	100
Duration (Hours)	2
Usable Energy (MWh)	200
90% Depth of Discharge Cycles/Day	1
Operating Days/Year	350
Charging Cost (\$/kWh)	\$0.033
Fixed O&M Cost (\$/kWh)	\$3.00
Fixed O&M Escalator (%)	2.5%
Charging Cost Escalator (%)	1.97%
Efficiency (%)	91%
Capital Structure	
Debt	20.0%
Cost of Debt	8.0%
Equity	20.0%
Cost of Equity	12.0%
Taxes	
Combined Tax Rate	40.2%
Economic Life (years)	20
MACRS (Year Schedule)	5 Years
Federal ITC - BESS	40%
Capex	
Total Initial Installed Cost (\$/kWh) ⁸	\$169
Extended Warranty (% of Capital Cost)	0.7%
Extended Warranty Start Year	3
Total Capex (\$m)	\$34

Source: Lazard estimates and publicly available information.

Note: Numbers presented for illustrative purposes only.

* Denotes unit conversion.

1 Assumes half-year convention for discounting purposes.

2 Total Generation reflects (Cycles) x (Available Capacity) x (Depth of Discharge) x (Duration). Note for the purpose of this analysis, Lazard accounts for degradation in the available capacity calculation.

3 Charging Cost reflects (Total Generation) / [(Efficiency) x (Charging Cost) x (1 + Charging Cost Escalator)].

4 O&M costs include general O&M (BESS plus any relevant Solar PV or Wind O&M, escalating annually at 2.5%), augmentation costs (incurred in years needed to maintain usable energy at original storage module cost) and warranty costs starting in year 3.

5 Reflects initial debt financing to fund capex.

6 Reflects initial cash outflow from equity sponsor.

7 Reflects a "key" subset of all assumptions for methodology and illustration purposes only. Does not reflect all assumptions.

8 Initial Installed Cost includes inverter cost, module cost, balance-of-system cost and EPC cost.

■ Technology-Dependent

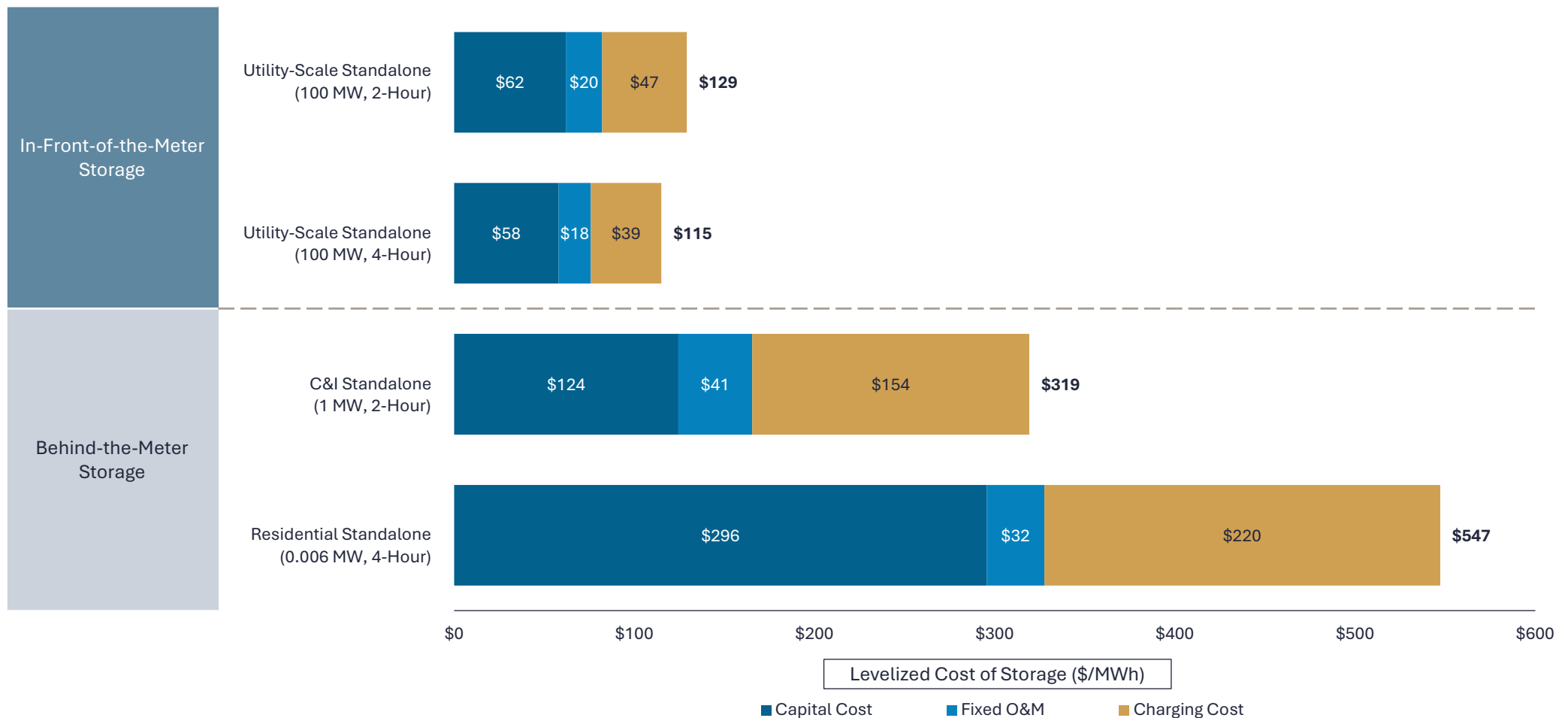
■ Consistent Across
Versions/Technologies

Levelized Cost of Storage—Key Assumptions

	Units	Utility-Scale Standalone						C&I Standalone			Residential Standalone		
		(100 MW/200 MWh)			(100 MW/400 MWh)			(1 MW/2 MWh)			(0.006 MW/0.025 MWh)		
Power Rating	MW	100			100			1			0.006		
Duration	Hours	2.0			4.0			2.0			4.2		
Usable Energy	MWh	200			400			2			0.025		
90% Depth of Discharge Cycles/Day	#	1			1			1			1		
Operating Days/Year	#	350			350			350			350		
Solar/Wind Capacity	MW	0.00			0.00			0.00			0.000		
Annual Solar/Wind Generation	MWh	0			0			0			0		
Project Life	Years	20			20			20			20		
Annual Storage Output	MWh	63,000			126,000			630			8		
Lifetime Storage Output	MWh	1,260,000			2,520,000			12,600			158		
Initial Capital Cost—DC	\$/kWh	\$113	—	\$244	\$107	—	\$232	\$238	—	\$445	\$721	—	\$1,338
Initial Capital Cost—AC	\$/kW	\$26	—	\$70	\$25	—	\$67	\$40	—	\$80	\$0	—	\$0
EPC Costs	\$/kWh	\$29	—	\$122	\$28	—	\$116	\$56	—	\$168	\$0	—	\$0
Solar/Wind Capital Cost	\$/kW	\$0	—	\$0	\$0	—	\$0	\$0	—	\$0	\$0	—	\$0
Total Initial Installed Cost	M \$	\$31	—	\$80	\$56	—	\$146	\$1	—	\$1	\$0	—	\$0
Storage O&M	\$/kWh	\$3.0	—	\$8.2	\$3.0	—	\$8.0	\$7.3	—	\$9.1	\$0.0	—	\$0.0
Extended Warranty Start	Year	3			3			3			3		
Warranty Expense % of Capital Costs	%	0.65%	—	1.50%	0.66%	—	1.85%	0.50%	—	1.30%	0.00%	—	0.00%
Investment Tax Credit (Solar)	%	0%			0%			0%			0%		
Investment Tax Credit (Storage)	%	30.00%	—	40.00%	30.00%	—	40.00%	30.00%	—	40.00%	30.00%	—	40.00%
Production Tax Credit	\$/MWh	\$0			\$0			\$0			\$0		
Charging Cost	\$/MWh	\$33			\$27			\$111			\$152		
Charging Cost Escalator	%	1.97%			1.97%			1.97%			1.97%		
Efficiency of Storage Technology	%	91%	—	87%	92%	—	86%	92%	—	88%	91%	—	88%
Unsubsidized LCOS	\$/MWh	\$129	—	\$277	\$115	—	\$254	\$319	—	\$506	\$547	—	\$860

Levelized Cost of Storage Components—Low End (\$/MWh)

Capital costs, fixed operating costs and charging costs contribute to the all-in cost in varying proportions depending on the specific energy storage use case and configuration



Levelized Cost of Storage Components—High End (\$/MWh)

Capital costs, fixed operating costs and charging costs contribute to the all-in cost in varying proportions depending on the specific energy storage use case and configuration



Energy Storage Use Cases—Overview

By identifying and evaluating selected energy storage applications, Lazard’s LCOS analyzes the cost of energy storage for in-front-of-the-meter and behind-the-meter use cases

		Use Case Description	Technologies Assessed
In-Front-of-the-Meter Storage	Utility-Scale Standalone	• Large-scale energy storage system designed for rapid start and precise following of dispatch signal • Variations in system discharge duration are designed to meet varying system needs (i.e., short-duration frequency regulation, longer-duration energy arbitrage¹ or capacity, etc.) – To better reflect current market trends, this analysis analyzes 2- and 4-hour durations²	• Lithium Iron Phosphate (LFP) • Lithium Nickel Manganese Cobalt Oxide (NMC)
	Commercial & Industrial Standalone	• Energy storage system designed for behind-the-meter peak shaving and demand charge reduction for C&I users – Units are often configured to support multiple commercial energy management strategies and provide optionality for the system to provide grid services to a utility or the wholesale market, as appropriate, in a given region	• Lithium Iron Phosphate (LFP) • Lithium Nickel Manganese Cobalt Oxide (NMC)
Behind-the-Meter Storage	Residential Standalone	• Energy storage system designed for behind-the-meter residential home use—provides backup power and power quality improvements – Depending on geography, can arbitrage residential time-of-use (“TOU”) rates and/or participate in utility demand response programs	• Lithium Iron Phosphate (LFP) • Lithium Nickel Manganese Cobalt Oxide (NMC)

Source: Lazard estimates and publicly available information.

1 For the purposes of this analysis, “energy arbitrage” in the context of storage systems paired with solar PV includes revenue streams associated with the sale of excess generation from the solar PV system, as appropriate, for a given use case.

2 The Value Snapshot Case Studies only evaluate the 4-hour utility-scale use case.

Energy Storage Use Cases—Illustrative Operational Parameters

Lazard's LCOS evaluates selected energy storage applications and use cases by identifying illustrative operational parameters ¹

- Energy storage systems may also be configured to support combined/“stacked” use cases

		A	B			C	B x C = D	E	F	D x E x F = G	A x G = H
		Project Life (Years)	Storage (MW) ²	Solar/Wind (MW)	Battery Degradation (per annum)	Storage Duration (Hours)	Nameplate Capacity (MWh) ³	90% DOD Cycles/Day ⁴	Days/Year ⁵	Annual MWh ⁶	Project MWh
In-Front-of-the-Meter Storage	Utility-Scale Standalone	20	100	–	2.6%	2	200	1	350	63,000	1,260,000
		20	100	–	2.6%	4	400	1	350	126,000	2,520,000
Behind-the-Meter Storage	Commercial & Industrial Standalone	20	1	–	2.6%	2	2	1	350	630	12,600
	Residential Standalone	20	0.006	–	1.9%	4	0.025	1	350	8	158

Source:

Note:

- Lazard estimates and publicly available information.
- Operational parameters presented herein are applied to Value Snapshot and LCOS calculations. Annual and Project MWh in the Value Snapshot analysis may vary from the representative project.
- The use cases herein represent illustrative current and contemplated energy storage applications.
- Indicates power rating of system (i.e., system size).
- Indicates total battery energy content on a single, 100% charge or “usable energy”. Usable energy divided by power rating (in MW) reflects hourly duration of system. This analysis reflects common practice in the market whereby batteries are upsized in year one to 110% of nameplate capacity (e.g., a 100 MWh battery actually begins project life with 110 MWh).
- “DOD” denotes depth of battery discharge (i.e., the percent of the battery’s energy content that is discharged). A 90% DOD indicates that a fully charged battery discharges 90% of its energy. To preserve battery longevity, this analysis assumes that the battery never charges over 95%, or discharges below 5%, of its usable energy.
- Indicates number of days of system operation per calendar year.
- Augmented to nameplate MWh capacity as needed to ensure usable energy is maintained at the nameplate capacity, based on Year 1 storage module cost.
- Usable energy indicates energy stored and available to be dispatched from the battery.

 = “Usable Energy”⁷

Lazard's LCOE+ will continue to evolve over time, and we appreciate that there can, and will be, varied views regarding the specifics of our analyses. Accordingly, we would be happy to discuss any of our underlying assumptions and analyses in further detail—and, to be clear, we welcome these discussions as we try to improve our studies over time. In that regard, the studies remain our attempt to contribute in a differentiated and impactful manner to the Industry.

More generally, Lazard remains committed to our Power, Energy & Infrastructure Group clients, who remain our highest priority. In that regard, we believe that we have the greatest allocation of resources and effort devoted to this sector of any investment bank. Further, we have an ongoing and intense focus on strategic issues that require long-term commitment and planning. Accordingly, Lazard strives to maintain its preeminent position as a thought leader and leading advisor to clients on their most important matters, especially in this Industry.

If you have any questions regarding this memorandum or Lazard's LCOE+, please feel free to contact any member of the Lazard Power, Energy & Infrastructure Group, including those listed below.

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